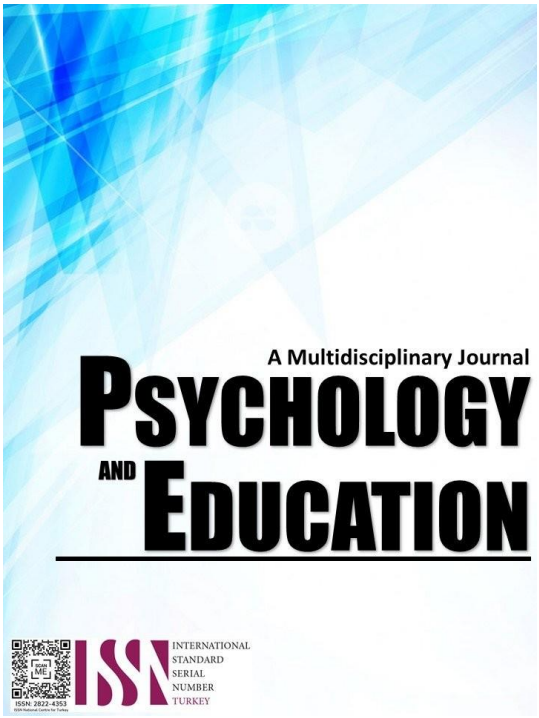
DIGITAL GAME-BASED LEARNING: ITS CONTRIBUTION TO NUMERACY SKILLS AMONG GRADE 1 LEARNERS



PSYCHOLOGY AND EDUCATION: A MULTIDISCIPLINARY JOURNAL

Volume: 34
Issue 5
Pages: 592-602
Document ID: 2025PEMJ3280
DOI: 10.70838/pemj.340504
Manuscript Accepted: 02-28-2025

Psych Educ, 2025, **34**(5): 592-602, Document ID:2025PEMJ3280, doi:10.70838/pemj.340504, ISSN 2822-4353

CC BY NC

Research Article

Digital Game-Based Learning: Its Contribution to Numeracy Skills among Grade 1 Learners

Aimee A. Buray,* Reynilda C. Alferez
For affiliations and correspondence, see the last page.

Abstract

The purpose of the study was to assessed the contributions of Digital Game-Based Learning to the numeracy skills among Grade 1 learners. It took place at Andales Integrated School, Initao North District, Division of Misamis Oriental during the Second Quarter of the School Year 2024-2025. Using a quasi-experimental design, 34 learners were treated using Digital Game-Based learning. The mean, standard deviation, and paired t-test were applied to the statement of the problem. With the Rapid Mathematics Assessment as a tool, the participants were tested before and after the treatment. Results indicated that there was a significant shift in scores from the pretest to the posttest, indicating a marked improvement in the numeracy skills of the first graders following the implementation of digital game-based learning strategies. This implied that adding digital games to classroom settings can promote a more engaging and successful instructional process. The study offered suggestions for integrating digital game-based learning in classroom settings and emphasized how it improved the performance of learners. As a result, school officials may keep looking for support to allocate funds for the school's technology upgrades.

Keywords: *digital game-based learning, pretest, posttest, quasi-experimental design, rapid mathematics assessment*

Introduction

At present, first-graders encountered many obstacles as they work to improve their numeracy abilities. A variety of factors, including individual learning variations, the school environment, instructional approaches, and developmental factors, might contribute to these issues. Thus, gaining a strong conceptual grasp of fundamental numeracy ideas like counting, recognizing numbers, and performing simple addition and subtraction arithmetic operations can be difficult. Understanding the significance and connections between numbers may be difficult for certain learners.

Learners started developing the foundational numeracy skills in Grade 1, which serve as the basis for more complex math ideas in later grades. These abilities included recognizing numbers, counting, comprehending quantities, and doing simple arithmetic operations like addition and subtraction. For daily chores like handling money, recognizing dimensions, determining the time, and performing simple computations, numeracy abilities were crucial. Learners frequently meet real-life circumstances that can be immediately applied to these practical skills. First-grade numeracy proficiency establishes a solid basis for future academic achievement. As they move through the educational system, it helped learners interact with arithmetic topics with greater confidence, which enhanced their overall academic performance.

Learners in Grade 1 were still working on their capacity to pay attention and concentrate for prolonged amounts of time. They might find it difficult to focus during numeracy exercises, which resulted in unfinished work or computation errors. Learners with varying learning needs and skills were commonly seen in first-grade classrooms. Uneven academic development and comprehension gaps might result from teachers' struggles to deliver differentiated education that was tailored to each learner's unique requirements.

Numerous government initiatives and reforms, such as teacher training programs, curriculum adjustments, and interventions to address learning gaps, have been implemented in an effort to increase numeracy abilities. Nonetheless, the effects of these programs may differ in various locales and educational institutions. The Guidelines on the Early Language, Literacy, and Numeracy Program, also known as Department Order Number 12 series 2015, serve as the foundation for one of the Department of Education's primary attempts to close the gaps in numeracy. The Department of Education's Early Language, Literacy, and Numeracy Program aimed to prepare all Filipino children for academic success and lifetime learning by making sure they develop the fundamentals of language, literacy, and numeracy early in their school careers, however, still encountered gaps.

On the other hand, using digital game-based learning to improve first-graders numeracy skills were entertaining and engaging given the state of technology. Due to the growing interest of children in playing video games, the usage of these games has become a hot topic. Sabirli and Coklar (2020), which investigated how educational games affect learners' academic achievement, found that incorporating digital instructional games into the curriculum improved learners accessed to the material and boosts their performance noticeably. A variety of subjects have been taught through digital games. Topics include language, social science, science, and nature, either social and soft skills (i.e., science, mathematics, or talents for teamwork) (Andersen & Rustad, 2022).

Hussein et al. (2022) stated that learning through digital games was a learner-centered method, where instructional objectives and materials were interwoven in gaming activities and inspired learners to study and advance their abilities and expertise by giving them fun and engaging experience surroundings for learning. Digital game-based learning will be employed to make sure that learners improved their numeracy abilities and grasp mathematical ideas. Therefore, this study aimed to determine the contribution of digital game-based learning to the numeracy skills of Grade 1 learners. This study was conducted in Andales Integrated School, Initao North

District, Division of Misamis Oriental, for the School Year 2024-2025.

The researcher has been a Grade 1 adviser for nearly two years, and she was excited to carry out this study to address learning gaps, particularly in numeracy. The researcher hoped that the study's findings greatly help first-grade learners by enhancing their numeracy abilities. This study thereby helped first-grade educators in providing timely, high-quality instruction that was precisely tailored to the requirements and interests of the learners in the first grade.

Research Questions

The study aimed to assess the contributions of digital game-based learning to the numeracy abilities of Grade 1 learners at Andales Integrated School, Initao North District, Division of Misamis Oriental, in the School Year 2024–2025. Thus, this study answered the following specific questions:

1. What are the pretest scores of the respondents?
2. What are the posttest scores of the respondents?
3. Is there a significant difference between the pretest and posttest scores of the respondents?
4. What action plan can be advised from the results of the study?

Methodology

Research Design

The research design used in this study was the one-group pretest and posttest research design. A pretest-posttest design involved measuring the dependent variable twice: once before the implementation of the treatment and once following it (Chiang et al., 2020). To employ this research design, the participants were given pretest to measure their prior knowledge. After the pretest, the researcher conducted the math lessons using the Digital Game-Based Learning. After the implementation, posttest was given to the participants to measure the effectiveness of the Digital Game-Based Learning.

Respondents

The participants of the study were the Grade 1 learners, which comprised one section with 57 learners of Andales Integrated School, Initao North District, Division of Misamis Oriental, during the second quarter of the School Year 2024-2025. With the learners' academic performance during the first quarter, the participants were chosen based on their academic performance in Mathematics. Those learners whose grades belong to 75-79 (fairly satisfactory) and 80-84 (satisfactory) ratings were the participants of the study. Overall, there were 34 participants in the study based on their academic performance in Mathematics. They had the pretest using the Rapid Mathematics Assessment, and they received Digital Game-Based Learning as treatment. According to Rivas et al. (2020), pretests gave researchers a baseline against which to compare the findings of posttests. After the treatment, they received a posttest.

Using a pretest as part of the sampling process provided the researcher with a special chance to improve the learner's sturdiness and comprehension of the target population. Pretesting provided valuable insights that immediately influenced the participants' selection process, resulting in a more rigorous and reliable research procedure.

Instrument

Based on DM-CT-2024, the Rapid Math Assessment, Comprehensive Rapid Literacy Assessment (CRLA) for Grades 1–3, and Philippine Informal Reading Inventory (Phil-IRI) Assessment for Grades 4–6 was administered nationwide by the Department of Education (DepEd) through the Curriculum and Teaching Strand.

One thorough and effective way to assess pupils' mathematical competency was through the use of Rapid Mathematics Assessment (RMA). It was a formative evaluation that gave teachers important information about how well their learners grasp mathematical ideas and abilities, empowering them to plan lessons and deliver interventions accordingly. Based on fundamental learning competencies, the rapid mathematics assessment was intended to pinpoint learners' areas of strength and growth in their mathematical aptitude. To gauge progress, there were two assessments: an oral pre-assessment at the beginning of a five-week program and a post-assessment at the conclusion. A teacher's guide, student materials, and automated scoring sheets are included in the assessment resources.

The resources for delivering the Grade 1 Rapid Mathematics Assessment were included in the booklet. It was ready to ensure that the required resources were accessible when the test was administered. Most of the tasks in this evaluation call for respondents to work with cutouts, pictures, and objects. The participants responded to the assessment tasks orally. On the materials, they did not write. For most assessment tasks, one set of materials might be adequate for every participant, although some might require copying and cutting. It's recommended to paste these materials on cardboard and use them again. The researcher was responsible for making sure all required materials were ready and for reading the Teacher's Booklet that goes with the test. It was anticipated that the researcher was acquainted with the resources, including how to use them and what to use them for.

The Rapid Mathematics Assessment process was child-centered. The researcher created a laid-back atmosphere and friendly relationship with the respondents. The researcher started the conduct of Rapid Mathematics Assessment, both pretest and posttest, by

having brief discussions with the participants about subjects they might find interesting. This took the researcher one to two minutes to complete. It was advantageous if participants viewed this assessment as more similar to a fun game or activity than a test that respondents fear. The researcher informed the participants that the evaluation did not function as a test and that the results did not impact their math grades. The participants did each job within the allotted time and had to provide their responses verbally. To determine when the time is up, a timer was employed. The researcher started the assessment as soon as the participant was prepared.

Procedure

The study commenced with the approved request letters from the Schools Division Superintendent, Public Schools District Supervisor, and School Principal. It took place in the second quarter of the School Year 2024-2025 at Andales Integrated School, Initao North District, Division of Misamis Oriental.

With the learners' academic performance during the first quarter, the participants were chosen based on their grades. Those learners who garnered 75-79 (fairly satisfactory) and 80-84 (satisfactory) grades were the participants of the study. They had the pretest using the Rapid Mathematics Assessment, and they received Digital Game-Based Learning as a treatment. After the presentation of lessons through Digital Game-Based Learning, a posttest was administered to determine any changes in the respondent`s knowledge about the lessons presented. Thus, the first quarter grades of the respondents have been evaluated to determine the participants of the study. Strict confidentiality has been emphasized in handling respondents` information. When the researcher has the data needed, the results were collated and submitted to the accredited Statistician of the school.

Data Analysis

To obtain the actual details required, the data were tallied and evaluated. The subsequent statistical instruments were utilized to address the various issues raised:

For Problems 1 and 2, Frequency and Percentage were used to determine the pretest and posttest scores of the participants.

For Problem 3, a Wilcoxon Signed-Rank Test was used to test the significant difference between the pretest and posttest scores of the participants.

Results and Discussion

Presentation, analysis, and interpretation of data were covered in this chapter. It presented the collected data, analysis of the statistical results, and analysis of the results. These were presented in tables following the sequence of the specific research question addressed by the application of Digital Game-Based Learning to Grade 1 learners.

Problem 1. What are the pretest scores of the participants?

Table 1 presented the pretest scores of the participants. The data revealed that the highest frequency count was found in the "Developing (Nearly Proficient)" category, with 21 participants (61.8%), while the lowest frequency count was in the "Emerging (Low Proficient)" category, with 13 participants (38.2%). This distribution indicated that while a significant number of participants were developing their numeracy skills, a notable proportion was still emerging, suggesting a need for targeted instructional strategies. The presence of learners in the "Emerging" category highlighted the importance of understanding the diverse proficiency levels within the classroom, which can inform the design of effective educational interventions.

Table 1. *Pretest Scores*

Score Range	*Frequency Count*	*Percentage (%)*	*Description*
1 – 8	0	0	Emerging (Not Proficient)
9 – 17	13	38.2	Emerging (Low Proficient)
18 – 26	21	61.8	Developing (Nearly Proficient)
27 – 29	0	0	Transitioning (Proficient)
30 -35	0	0	At Grade Level (Highly Proficient)
Total	34	100.0	

The same findings were obtained by Tallud and Caballes (2023), who assessed the numeracy proficiency of learners from Jose Magsaysay Elementary School using a modified local assessment tool. They found that a sizable portion of the learners were not reaching the required competency levels indicating serious gaps in numeracy. According to Bezuidenhout (2020), these findings have important implications as well. They suggested that early intervention may harm learners as they advance through the academic program year by year, ultimately resulting in long-term academic difficulties. The numeracy skills of first-graders still have much room for improvement, according to these findings. One possible tactic to support this development was use of digital game-based learning (DGBL).

There was proof that using games can significantly increase learners' motivation and engagement, two critical elements that have wide-ranging effects on numeracy development. For example, Rohendis (2019), game-assisted multimedia intervention for mathematics learning, reported greater interest and engagement in the adventure games vocabulary learning and problem-solving tasks, which led

to improved mathematics attainment. According to the results of one study, supplementing educational media with locally sourced elements improved learners' numeracy skills, and culturally rich content could increase pedagogical efficacy (Hidayah et al., 2021).

As the paper notes, engaging and dynamic learning environments are also essential for the success of numeracy instruction. Learners who employed these strategies outperformed those who used more traditional approaches in terms of numeracy on problem-based learning with an ethnomathematics twist (Utama, 2023). It begs the question of whether employing interactive and culturally appropriate teaching strategies like those found in video games can significantly impact learners' early numeracy development. Developing instructional materials that incorporate problem- and project-based learning can greatly boost learners' creativity and motivation for learning, which are critical qualities for academic success (Sutama et al., 2023).

This was especially crucial in the Philippine context, where putting instructional strategies into practice that motivate learners to actively study could improve their academic performance. Consequently, learning through Digital Game-Based Learning not only increased enjoyment but also gradually improved academic performance. The results of the pretest indicated a statistically significant distribution of participants' abilities, which indicated that the percentage of participants in both the developing and emerging categories was significant. This showed receptivity to particular interventions like Digital Game-Based Learning, which could help people learn numeracy skills. The previous research was corrected when it suggested that a few practical learning techniques that were supported by pretest results can help learners learn much more effectively.

Problem 2. What are the posttest scores of the participants?

Table 2 displayed the posttest scores of the participants after the integration of digital game-based learning into the class session. The data revealed that the highest frequency count was found in the "Developing (Nearly Proficient)" category, with 27 participants (79.4%), while the lowest frequency count was in the "Emerging (Low Proficient)" category, with only 4 participants (11.8%). This significant shift in scores from the pretest to the posttest indicated a marked improvement in the numeracy skills of the first graders following the implementation of digital game-based learning strategies. The absence of participants in the "Emerging (Not Proficient)" and "At Grade Level (Highly Proficient)" categories further emphasized the effectiveness of the intervention, suggesting that the majority of learners have moved toward a more proficient understanding of numeracy concepts.

Table 2. *Posttest Scores*

Score Range	*Frequency Count*	*Percentage (%)*	*Description*
1 – 9	0	0	Emerging (Not Proficient)
10 – 17	4	11.8	Emerging (Low Proficient)
18 – 26	27	79.4	Developing (Nearly Proficient)
27 – 29	3	8.8	Transitioning (Proficient)
30 -35	0	0	At Grade Level (Highly Proficient)
Total	34	100.0	

The significance of all data cannot be overstated, particularly in shaping educational strategies to enhance learners' numeracy skills. The predominance of participants in the "Developing" category highlights that incorporating Digital Game-Based Learning (DGBL) was a compelling approach. It not only captured learners' interest but also actively involved them in their educational journey. Evidence consistently supported the effectiveness of game-based learning in boosting academic outcomes. For instance, research by Sáez-López et al. (2022), underscored how gamification markedly enhanced learners' participation and academic results in elementary education, underscoring the advantages of interactive learning environments. These findings aligned with the current study's conclusions, which demonstrated that digital game-based learning was a fresh, dynamic, and highly effective pedagogical method.

Moreover, existing research highlighted the critical role of interactive and engaging learning settings in developing numeracy skills. On the other hand, Patmanthara et al. (2019) showed that using educational games can significantly enhance learning outcomes across various disciplines, including mathematics. This evidence supported the idea that incorporating game-based strategies into educational programs can deepen learners' comprehension and practical application of mathematical concepts. The improved performance in post-test results further validated the potential of technology integration in creating more dynamic and impactful learning experiences.

Aligned with this perspective, the posttest findings emphasized the critical need for continuous assessment to evaluate the success of educational initiatives. Gu et al. (2021) explored how assessment plays a pivotal role in pinpointing areas that require attention and in customizing teaching approaches to address the unique needs of learners. In this study, the pretest functioned as an initial benchmark, enabling educators to assess the effect of the DGBL strategy on student achievement. The findings highlighted the necessity of regular assessments to guide teaching methods and ensure that learners are making substantial and meaningful progress.

This was particularly shown to help the enhancement of numeracy in elementary learners like Grade 1 students from a study of Digital Game-Based Learning (DGBL). A systematic review of the literature on Digital Game-Based Learning identified several studies indicating the benefits of Digital Game-Based Learning among learners in relation to their mathematical abilities. A specific study, for example, by Sabirli and Çoklar (2020), investigated the influenced of educational digital games on the motivation of some elementary school learners as well as their learning attitudes. Results showed that learners exposed to a digital game exhibited greater problem-solving ability and more favorable attitudes toward mathematics than learners who were not exposed to the digital game supporting

this assertion that digital game-based learning can be effectively utilized to improve numeracy skills in young children. This fits in with the broader literature that highlighted exciting educational tools as one strategy for developing a love for learning and academic achievement since they may promote a love of learning.

Also, Hamidah et al. and Gokhan (2023) studied Wordwall as a game-based learning interactive media and found that game-based learning positively affected the literacy and numeracy of elementary school learners. Interactive digital tools have been found to foster responsive classrooms with opportunities for skill development, a finding emphasized in their research. This was especially true for the Philippines because there was a need for innovative ways of teaching to meet diverse learning needs.

Problem 3. Is there a significant difference in the posttest and pretest scores of the participants?

Table 3 presented the difference between the posttest and pretest scores of the participants. The data revealed that the posttest scores had a highly significant difference between the posttest scores of the participants. Thus, the null hypothesis, which stated that there was no significant difference between the posttest and pretest scores of the participants, was rejected. The results suggested that the Digital Game-Based Learning (DGBL) intervention has had a substantial positive impact on the participants' numeracy skills (Anwar et al., 2022).

Table 3. *Difference1 Posttest and Pretest Scores*

		Static	*Df*	*p-value*	*Mean Difference*	*SE Difference*
Posttest	Pretest	4.4	33.0	<0.001	0.353	0.0832

*Note: 2 – Wilcoxon Signed-Rank Test ** - p < 0.01 *** - p < 0.001 – p > 0.05 * - p < 0.05*

These findings have important implications for educational practices aimed at enhancing the numeracy skills of young children. The enhancement witnessed in post-test scores indicated that digital game-based learning might be beneficial in learners' engagement and in making mathematical concepts comprehensible. This was consistent with a recent emerging body of research that supported the efficacy of game-based learning in classrooms.

These findings significantly affected the teaching strategies for enhancing young pupils' arithmetic abilities. Digital game-based learning was one approach that would maintain learners' interest and even support their understanding of mathematical concepts, as seen from improved exam scores. This is related to the growing body of research that confirmed games as an appropriate addition to the classroom. For example, Sutama et al. (2023), found out that learners enhanced their arithmetic aptitude after making use of new learning resources like games. Learners usually do better if one is teaching by being fun and experiential, they added.

In addition, the results were usually consistent with Dewi's research, which revealed that the presence of games in learning motivated and improved the overall learning of learners (Dewi, 2023). This was due to the fact that Digital Game-Based Learning was able to develop an interesting and interactive learning environment, which served as the foundation and was of paramount importance to young learners, whose learning was enhanced by experiential and engaging learning experiences. This can be attributed to the constructivist concept that learning occurs through interactive and active means, such as engaging with the environment.

Additionally, Mukh et al. (2021) provided insights into the effectiveness of digital games in the learning process through their study on the Digital Learning Games Scale (DLGS). Their research, which involved teachers from elementary schools, found that digital games significantly contributed to enhancing learners' learning experiences and outcomes. This underscored the importance of teacher training and support in successfully implementing Digital Game-Based Learning strategies.

Thus, a study by Travinio and Gonzales (2022), investigated the use of game-based applications to improve learners' performance and engagement in the Philippine environment. This was consistent with the findings of the current study, which showed that post-test scores significantly improved when Digital Game-Based Learning was incorporated.

Problem 4. What action plan can be designed based on the results of the study?

I. Rationale

Numeracy in early childhood education served as the foundations for later school achievement, including, but not limited to, problem-solving abilities. Example, counting, addition, and subtraction were some of the most important numeracy skills for a child's first-grade cognitive and academic progress (LC). Unfortunately, there were a lot of us smart young learners who, because of inadequately pedagogical content, do not wished to learn such fundamental stuff due to lack of interest and engagement and because they were not compelling and experiential.

As a result of the increasing digital world, there was a tendency to apply Digital Game-Based Learning (DGBL) in a practical form, by using it to teach numeracy in a classroom environment. Several studies suggested that interactive computer games have the ability to produce, facilitate, and deliver relevant feedback to young learners. Hence, the learners learned the knowledge at the speed that they judged to be appropriate for them, as dictated by the individual differences in the learning speed the students exhibit. Another effective way that gaming can help overcome "mathemaphobia" was by either promoting a sense of mastery over the topic or by making mathematics appealing and preventing a bad experience.

The application of Digital Game-Based Learning in fields such as mathematics offered an opportunity to transform educational approaches in the Philippines, where classroom materials may be lacking and traditional teaching was not always effective in reaching all children. No matter what baseline in numeracy levels the learner starts with, the tools that can support the education towards building a good numeracy foundation were available for the teacher to use, by integrating technology into the teaching process. problem-solving, through Digital Game-Based Learning.

Aiming to enhance numeracy skills, this action plan intended to introduce digital game-based learning tools to the Grade 1 classroom. The strategy was designed to increase student motivation, engagement, and learning results in a stimulating and effective way by introducing interactive, game-like math learning activities.

II. Objectives

The major aimed of this action plan were to tackle current numeracy challenges faced by year-one children and to the benefits of including digital game-based learning in the curriculum:

1. Before digital games-based learning was available, a pretest was employed to measure the degree of numeracy skill of the first-graders, especially regarding concepts of counting, number sense and additive/subtractive computation of their grade level., number identification, and shallow additive and subtractive operation.

2. To provide interactive games that supported the recall of key numeracy concepts in everyday classroom mathematics by utilizing digital game-based learning tools. In order to schedule at least 10 to 15 min of a digital game session per lesson, emphasis was given to elementary concepts (e.g., simple addition, subtraction, and even ordinal number recognition).

3. Employing post-tests and game progress logs from digital applications, conduct periodic progress assessments to track and evaluate the effect of digital game-based learning on numeracy skills in order to determine whether the use of digital game-based learning has affected students' numeracy levels.

4. For the sake of giving parents an opportunity to further develop the numeracy of their children beyond the classroom and reinforced the numeracy skills they have learned in school; we briefly wanted to show the ways in which the use of digital game based-learning apps can be taught by parents in and out of the classroom.

5. For providing an intensive learning environment, for instance, in which children were motivated and enthusiastic to learn math using digital games and acknowledged (e.g., digital badges, stickers, progress reports) fulfilling achievements with rewards.

6. To ensure the longevity of the program by providing teachers with the resources and guidance they needed to continue using digital game-based learning in their classrooms and explored the funding avenues for the development of digital resources).

Conclusions

This study assessed the contribution of digital game-based learning on Grade 1 learners' numeracy skills in the school year 2024-2025 at Andales Integrated School, Initao North District, Division of Misamis Oriental. The results indicated that DGBL could be employed as a useful method to enhance the development of learners' numeracy skills, considering the vast improvement in the post-test score upon its implementation. The majority of learners showed noticeable progress from their pretest to posttest results, with a shift towards higher proficiency categories. This suggested that digital games, when integrated effectively into the learning process, can enhance learner engagement, understanding, and application of mathematical concepts.

Overall, the large variation between pretest and posttest scores, as established by the Wilcoxon Signed-Rank Test, represented how actual Digital Game-Based Learning really was in its approach to teaching numeracy development. However, while the learners have made measurable advancements, some were still not at the "At Grade Level" standard. This meant that more support and strategies may need to be employed to ensure that all learners achieve full proficiency. In so doing, the importance of using technology in education lies more so in early grade levels, where it fosters a fun and active learning environment that encourages active participation and building foundational skills.

This study highlighted the potential of digital game-based learning in revolutionizing traditional pedagogy, thereby affording more immersive and personalized learning. It also opens up worthwhile insights to educators, school administrators, and policymakers on the actual teaching-learning advantages that could be reaped by simply tapping into educational technology. The findings of this study suggested that involving digital games in the classroom improved the numeracy skills of first-grade learners as well as their academic performance. Now, since schools are embracing advancements in technology, it became even more important to discuss other ways digital game-based learning can be developed and used to accomplish the task of developing different abilities in learners.

Based on the findings of this study, the following recommendations are proposed:

The entry of digital game-based learning to the Grade 1 curriculum in Mathematics may be recommended for enhancing numeracy skills among school heads. School districts may conduct professional development activities on the part of teachers that would enable the implementation, in an age-specific manner, of relevant digital game-based learning stools like I Know It, Mindly, Education.com,

Math Games, Math Play Ground, and Top Marks. There shall also be digital learning stations, either with tablets or computers. These will be available for small group rotations to ensure that all learners have constant access to these educational resources. To continue the use of digital game-based learning in a nondenominational way across different subjects, schools should develop a common digital pool of lesson plans, teaching strategies, and other resources. It will also be essential to explore obtaining funding for the program from external funding or grants to expand technology and digital tools access. Collaboration among teachers and the sharing of best practices will also help to continue developing and perfecting the program.

For teachers, providing consistent and structured opportunities for learners to practice numeracy skills through digital game-based learning should be a key strategy in the classroom. Teachers should be able to add 10-15 minutes of digital game sessions to their daily lessons, providing instant feedback for learners. Recording learners' performance through game progress reports will enable the identification of areas where individual learners may need additional support, thereby conducting targeted interventions to address specific learning needs. Student progress assessment is essential to monitor whether the digital game-based learning intervention improves learning outcomes. The administration of pre- and post-assessments to measure the learning effects of students in numeracy areas is highly recommended. Furthermore, by reviewing the game progress reports, teachers can continuously monitor each student's performance, so adjustments on the instructional approach may be made appropriately.

For parents, facilitating more involvement in the learning process will help solidify numeracy skills at home. To this end, schools will educate parents on the values of DGBL and how to assist their children in digitalized ways of their learning. This can be further enhanced by the following: a) printed guides, b) recommended educational apps, and c) online platforms created for communication and feedback (for example, Facebook groups).

For learners, digital game-based learning can impart numeracy skills innovatively and with fun. Including games in their study routine may have an impact on improving your confidence, allowing you to gain an enhanced comprehension of mathematical ideas, and sharpening your problem-solving skills. You can maintain their motivation by keeping a check on their development and exercising patience while overcoming various obstacles and levels. The secret is that one should use games for additional general mathematical development and, of course, to seek out ones that are at the same time entertaining and instructive.

Digital game-based learning needs to be communicated to the community, which includes game creators and neighborhood organizations, so they understand their role in the development of numeracy. Communities could provide a lighthearted, hands-on, and time-effective means by which people of all ages may develop their numeracy skills through collective efforts at involving, promoting, and enhancing the use of learning games. Whether in a classroom, home, or community center, digital game-based learning offers a special opportunity to make math learning enjoyable and effective. Future researchers can tap promising opportunities in future studies into fields like digital game-based learning and the development of numeracy skills. Filling the gaps in the current literature on the topic and further inquiry into new technologies, differences among various learner demographics, and the cognitive processes underlying Digital Game-Based Learning are key ways to enhance creating and utilizing educational games. Empirical research can add to our understanding of the impact of Digital Game-Based Learning on numeracy abilities and further develop a use for such to assist learners in different communities and situations.

References

Akram, H., & Abdelrady, A. H. (2023). Application of ClassPoint tool in reducing EFL learners' test anxiety: an empirical evidence from Saudi Arabia. Journal of Computers in Education, 10(3), 529–547. https://doi.org/10.1007/s40692-023-00265-z

Andersen, R., & Rustad, M. (2022). Using Minecraft as an educational tool for supporting collaboration as a 21st-century skill. Computers and Education Open, 3, 100094.

Anderson, A., & Liu, Y. (2021). Barriers to adoption of digital game-based learning. International Journal of Game-Based Learning, 11(3), 1-18.

Anwar, M., Suhartono, T., & Suryanto, T. (2022) Development of synectic learning model to improve explanation text writing ability. *EAI Conference Proceedings*. https://doi.org/10.4108/eai.20-9-2022.2324704

Asian Development Bank. (2020). Philippines: Education Sector Assessment, Strategy, and Road Map. Retrieved from https://www.adb.org

Bezuidenhout, H. S. (2020). The interface between early numeracy, language and learning environments: pedagogical implications. South African Journal of Childhood Education, 10(1). https://doi.org/10.4102/sajce.v10i1.923

Brau, A. (2020). Constructivism: A Theoretical Review of Learning. International Journal for Research in Education, 2(1), 1-10.

Chen, C. H., & Tu, H. T. (2021). Digital game-based learning in higher education: A quasi-experimental study of learning achievement, cognitive load, motivation, and flow.

Cherry, K. (2022). Experiential Learning Theory. Verywell Mind. https://www.verywellmind.com/experiential-learning-theory-2795435

Chiang, S. T., Lee, V., & Yen, C. F. (2020). Development and validation of the Chinese version of the adolescent health risk behaviors assessment. Journal of School Health, 90(11), 843-854.

Chi, J. (2023). Recent results of the Program for International Student Assessment (PISA) reveal poor performance of Filipino learners in mathematics, reading, and science. Journal of Educational Research and Development, 24(2), 123-130.

Chiu, L. (2022). The role of active learning in facilitating deeper conceptual understanding. Journal of Education and Learning, 12(2), 123-134.

Clements, D. H., & Sarama, J. (2019). Early childhood mathematics education research: Learning trajectories for young children. New York: Routledge.

Dabbous, O., Nouri, N., & Al-Khraishi, M. (2022). Game-Based Learning in Science Education: A Systematic Review. Sustainability, 14(3), 1429.

De Freitas, S., & Oliver, M. (2019). The cost of game-based learning: A systematic review. British Journal of Educational Technology, 50(4), 1524-1540.

Deabanico, G. J., & Caballes, D. G. (2023). Revolutionizing Learning: How eGamification Assessment is Changing Teaching? Zenodo (CERN European Organization for Nuclear Research). https://doi.org/10.5281/zenodo.8168422

Dede, C., & Richards, J. (2019). Collaborative Learning with Digital Games: A Review of the Literature. *Journal of Educational Computing Research*, 57(4), 550-569

Dettori, J. R., Norvell, D. C., & Chapman, J. R. (2019). Grab control! Choosing the right comparison group in an observational study. Global Spine Journal, 9(4), 456–458. https://doi.org/10.1177/2192568219847206

Dewi, B. I. P. (2023). Implementation of the class 6 teaching campus program to improve students' literacy and numeracy abilities at smpn 2 lenek. Polyhedron International Journal in Mathematics Education, 1(2), 94-105. https://doi.org/10.59965/pijme.v1i2.49

Dizon, J. (2024). Rapid Mathematics Assessment. Scribd Inc.https://www.scribd.com/document/700774539/RapidMathematicsAssessment

Emperador, MN. (2020). Claveria Approach Through Multime-Aided Teaching: Its Contribution to Reading Skills among Grade 1 Learners [A Master's thesis study, St. Peter's College]

Fitriana, N. (2023). Increasing Student Activeness Through Classpoint Percentage Media And Educational Games (Quizizz & Kahoot) In Chemistry Learning. Action: Journal of Classroom and School Action Research Innovation, 3(1), 35–41.

Galvez, D. (2019). Poor PISA scores a wake up call for PH education system- DepEd Chief. Inquirer.net. https://newsinfo.inquirer.net/1199952/fwd-briones-low-ph-pisa-results-a-wake-up-call#ixzz8fZFdfg4h

Garcia, E. (2020). Learning styles and student success. Educational Review, 23(4), 123-145.

Gigante, N. (2020). Numeracy among Filipino learners. Sun.star Pampanga. https://www.pressreader.com/philippines/sunstarpampanga/20201016/281625307780356

Ginsburg, H. P. (2021). The development of mathematical thinking. New York:Routledge.

Go, M. B., A, G. R., Velos, S. P., Dayupay, J., Dionaldo, W., Cababat, F., Bongo, M. F., Troussas, C., & Ocampo, L. (2022). Evaluating digital mathematical games in improving the basic mathematical skills of university students. International Journal of Mathematical Education in Science and Technology. Advance online publication. https://doi.org/10.1080/0020739X.2022.2089604

Gu, L., Zhang, W., & Li, H. (2021). Estimating difference-score reliability in pretest–posttest settings. *Journal of Educational and Behavioral Statistics*, 46(2), 1-20. https://doi.org/10.3102/1076998620986948

Hamidah, F., Setiawan, F., & Mirnawati, L. B. (2023). Strenghthening digital literacy of elementary school students through utilization of wordwall as game-based learning interactive media. Jurnal Ilmiah Sekolah Dasar, 7(2), 215-223. https://doi.org/10.23887/jisd.v7i2.55807

Heng, C. S. & Saed, S. A. R. (2020). Digital game-based learning (DGBL): An overview of studies, learning domains, and digital game-based learning types. International Journal of Information and Education Technology, 10(3), 176-181.

Hidayah, I. R., Kusmayadi, T. A., & Fitriana, L. (2021). E-book based on local wisdom to improve students' numeracy skill: is it effective?. Advances in Social Science, Education and Humanities Research. https://doi.org/10.2991/assehr.k.211122.020

Holbrey, C. E. (2020). Kahoot! Using a game-based approach to blended learning to support effective learning environments and student engagement in traditional lecture theatres. Technology, Pedagogy and Education, 29(2), 191-202.Available at: https://doi.org/10.1080/1475939X.2020.1737568.

Huang, Y., & Liao, C. (2023). The Impact of Digital Games on Cognitive Skills Development in Elementary Students. *Computers in Human Behavior*, 137, 107162.

Hussein, M. H., Al-Rawashdeh, A. H., & Al-Qadi, O. A. (2022). The Use of Digital Games in Education: A Comprehensive Review. International Journal of Interactive Mobile Technologies , 16(15), 91-106.

Irina, K. (2023). The Impact of Educational Game-Based Learning on Student Motivation: A Comprehensive Investigation. Journal of Educational Technology, 45(2), 120-145.

Jones, D., Smith, P., & Brown, L. (2019). The role of digital games in reducing math anxiety in young learners. *International Journal of Game-Based Learning*, 9(2), 1-16.

Kangan Institute, (2021). The importance of literacy and numeracy skills. Kangan Institute Blog. https://www.kangan.edu.au/study-with-us/our-blog/the-importance-of-literacy-and-numeracy-skills

Kaur, R., & Kaur, G. (2022). Using digital game-based learning to support students with learning difficulties in mathematics. *International Journal of Educational Research*, *54*, 101-112.

Khuda, B., Muhammad, H., Shumaila, S., Bushra, N., & Muhammad, F. F. (2022). Effectiveness of digital game-based learning strategy in higher educational perspectives. Journal of Education and e-Learning Research, 9(4), 258-268. 10.20448/jeelr.v9i4.4247

Kim, M.K., & Chang, K. (2020). The Effects of Digital Game-Based Learning on Academic Performance: A Meta-Analysis. *Computers & Education*, 144, 103703.

Lee, H., Kim, J., & Park, S. (2021). The effectiveness of a digital game intervention on number sense development in first-grade students. *Journal of Early Childhood Education*, 49(3), 221-234.

Lee, J. C., & Campbell, J. P. (2020). Digital game-based learning: Not the panacea for every learning objective. Educational Technology Research and Development, 68(4), 1283-1305.

Limniou Haratsis, A. (2021). Collaborative learning and peer interaction in higher education: A literature review. Procedia Computer Science, 183, 130-136.

Lorenzo-Alvarez, R., Rudolphi-Solero, T., Ruiz-Gomez, M. J., & Sendra-Portero, F. (2020). Game-based learning in virtual worlds: A multiuser online game for medical undergraduate radiology education within second life. Anatomical Sciences Education, 13(5), 602-617.Available at: https://doi.org/10.1002/ase.1927.

Mandinach, E. B., & Fisher, M. G. (2019). Digital game-based learning for middle school social studies: Exploring the impact on student learning. Contemporary Issues in Technology and Teacher Education, 19(3), 630-654.

Means, B., & Krause, K. L. (2021). Digital game-based learning: A systematic review. Journal of Computer Assisted Learning, 37(5), 456-479.

Minkiewicz, J., & Mejia, A. (2022). The ethics of data collection in digital game-based learning. International Journal of Game-Based Learning, 12(1), 1-20.

Mix, K. S., Huttenlocher, J., & Levine, S. C. (2021). Spatial Reasoning and Early.Math Abilities: A Meta-Analysis. Child Development, 92(5), 1681-1699.

Moral-Sanchez, S. N. (2021). Game-based learning and Breakout as comprehension techniques in solving math problems in primary education. Universidad De Málaga. https://hdl.handle.net/10630/22556

Morata, Jan Angelo & Morata, G & Caballes, Dennis. (2024). The use of digital game-based learning on numeracy of grade 7 students using class point: an input to professional development of teachers. 177-188.

Mukh, Y. N. A., Hashaikeh, S. A., & Abd-Rabo, A. M. (2021). Digital learning games scale (dlgs): a scale development study. International Journal of Emerging Technologies in Learning (iJET), 16(11), 140. https://doi.org/10.3991/ijet.v16i11.20709

Nasir, N. A., Nawawi, A., & Ismail, M. A. (2021). The accessibility of digital game-based learning: A systematic review. International Journal of Educational Technology in Higher Education, 18(1), 1-15.

National Center for Education Statistics. (2023). *National Center for EducationStatistics*. Retrieved from https://nces.ed.gov/

Ngoc Dan, T., Nguyen, H. T., & Tran, T. T. (2024). Research issues in elementary mathematics core areas: A review of techniques and gaming modes. Journal of Mathematics Education, 25(3), 456-468.

Noroozi, O., Mulder, R. H., Biemans, H. J. A., & Tormey, R. (2020). Effects of applying game-based learning approach on students' problem-solving skills and engagement. British Journal of Educational Technology, 51(6), 1921-1938.

Obenza-Tanudtanud, Dianne & Obenza, Brandon. (2024). Assessment of Educational Digital Game-Based Learning and Academic Performance of Grade Six Pupils. 3. 1-9. 10.54536/ajiri.v3i1.2338.

O'Brien, H. L., & Barton, M. (2020). The impact of digital game-based learning on student engagement and social interaction. Journal of Educational Technology Systems, 48(4), 496-517.

OECD. (2019). PISA 2018 Results (Volume I): What Students Know and Can Do. Retrieved from https://www.oecd.org/pisa/publications/pisa-2018-results-volume-I.htm

OECD. (2022). Teachers and Digital Technologies: A Global Perspective. https://www.oecd.org/education/teachers-and-digital-technologies-aglobal-perspective.htm

Owston, R., Wideman, H., & Park, G. (2020). Technical challenges in the implementation of game-based learning. Educational Technology Research and Development, 68(2), 641-654.

Patmanthara, S., Yuliana, O. D., Dwiyanto, F. A., & Wibawa, A. P. (2019). The use of ladder snake games to improve learning outcomes in computer networking. International Journal of Emerging Technologies in Learning (iJET), 14(21), 243. https://doi.org/10.3991/ijet.v14i21.10953

Philippine Statistics Authority. (2021). Philippine Education: Analytical Reports. https://psa.gov.ph

Prensky, M. (2019). *Digital Game-Based Learning*. Routledge.

Project Management Institute. (2020). A Guide to the Project Management Body of Knowledge (PMBOK® Guide) – Sixth Edition. Newtown Square, PA: Project Management Institute, Inc.

Raudenbush, S. W., & Willms, J. D. (2020). How Value-Added Measures of Teacher Effectiveness Can and Cannot Be Constructed from Test Score Data. Educational Researcher, 49(5), 302-315. ResearchGate.https://www.researchgate.net/publication/36

Reyes, F. J. F. D., & Caballes, D. G. (2022). Digital technology as a mode of instruction: An analysis. ResearchGate.https://www.researchgate.net/publication/36

Rivas, J. J., Arteaga, A. L., & López, M. I. (2020). The impact of pretests and posttests on research outcomes. Journal of Applied Research in Higher Education, 12(2), 202-215.

Rohendis, I. (2019). Game-assisted multimedia intervention for mathematics learning. Journal of Research in Mathematics Education, 50(4), 425-440.

Rosas, R., Nussbaum, M., & Heffernan, N. (2020). Digital game-based learning for mathematics: A meta-analysis. *Computers & Education*, *153*, 103809.

Roschelle, J., Pea, R., Hoadley, C.M., Gordin, D.N., Henry, L., & Means, B. (2020). The Future of Learning: A Research Agenda for the Next Decade. Educational Researcher, 49(7), 402-417.

Sabirli, Z. E. and Çoklar, A. N. (2020). The effect of educational digital games on education, motivation and attitudes of elementary school students against course access. World Journal on Educational Technology: Current Issues, 12(4), 325-338. https://doi.org/10.18844/wjet.v12i3.4993

Sabourin, J., & Lester, J. (2019). Game-based learning: A pedagogical Perspective.In Digital Game-Based Learning (pp. 3-18). Springer.

Sáez-López, J., Vázquez-Cano, E., Cadavieco, J. F., & López-Meneses, E. (2022). Gamification and gaming proposals, teachers' perceptions and practices in primary education. Interaction Design and Architecture(s), (53), 213-229. https://doi.org/10.55612/s-5002-053-011

Serrano, Katrina, "The effect of digital game-based learning on student learning: A literature review" (2020). Graduate Research Papers. 943. https://scholarworks.uni.edu/grp/943

Smith, M., Johnson, D., & Williams, R. (2020). The influence of digital games on motivation and engagement in mathematics learning for first-grade students. *Journal of Educational Psychology*, 112(5), 942-956.

Squire, K. (2021). *Video Games and Learning*. MIT Press.

Sun & Chen, (2021). Digital Game-based Pedagogical Activities in Primary Education: A Review of Ten Years' Studies. International Journal of Technology in Teaching and Learning. 16. 78-92. 10.37120/ijttl.2020.16.2.02.

Sutama, S., Asta, W. G., Purnomo, D. A., & Utomo, M. A. (2023). The effect of Problem-Based Learning (PBL) and Project-Based Learning (PjBL) on student's creativity and motivation. Jurnal Pendidikan MIPA, 24(2), 283–293. https://doi.org/10.23960/jpmipa/v24i2.2023.2840

Tallud, N. S. M. & Caballes, D. G. (2023). Bridging the gap of Jose Magsaysay Elementary school learners in numeracy: Makati localized numeracy assessment tool for stage 1 (grades 1-3). World Journal of Advanced Research and Reviews, 17(2), 678-685. https://doi.org/10.30574/wjarr.2023.17.2.0289

Travinio, R. D. J. & Gonzales, M. A. S. (2022). Game-based applications to enhance students' engagement and performance. International Journal of Research Publications, 103(1). https://doi.org/10.47119/ijrp1001031620223484

UNESCO. (2020). COVID-19 and Education: A Global Perspective. https://unesdoc.unesco.org/ark:/48223/pf0000373490

UNESCO. (2020). Education Sector Analysis: Philippines. Retrieved from https://en.unesco.org

Utama, F., Zaenuri, Z., & Pranoto, Y. K. s. (2023). Students numerical literacy ability in problem based learning with ethnomatematics nuances. Prima: Jurnal Pendidikan Matematika, 7(2), 98. https://doi.org/10.31000/prima.v7i2.8387

Van Eck, R., & Dempsey, J. V. (2020). *Educational Technology: Research and Practice*. Pearson Education.

Waltz-Budert, T. (2023). Pretest-Posttest Design | Definition, Types Examples.Study.com.https://study.com/learn/lesson/pretest-posttest

World Bank. (2021). The Digital Divide in Education. https://www.worldbank.org/en/topic/education/brief/the-digital-divide-in-education

Wu, H., Chang, K., & Kim, M.K. (2021). The Role of Motivation in Digital Game-Based Learning. *Educational Psychology Review*, 33(2), 415-438.

Yang, L., Chen, S., & Zhang, W. (2023). The effect of a game-based program on problem-solving abilities in Grade 1 students. *International Journal of Educational Technology in Higher Education*, 20(1), 1-15.

Affiliations and Corresponding Information

Aimee A. Buray
Andales Integrated School
Department of Education – Philippines

Reynilda C. Alferez, PhD
St. Peter's College – Philippines